Deliverable 5: NoSQL Document Store, Aggregations & Polyglot Integration

Project: Intelligence Fusion Platform • **Course:** DBMS Mini Project (UE24CS395)
Authors: Saatvik Gupta (PES1UG24CS395) & Satyam Goyal (PES1UG24CS424) • PES University (2026–2027)

1. Technology Justification: Relational SQL vs Document NoSQL

PostgreSQL: Enforces strict relational integrity, ACID transactional guarantees, foreign key cascades, and check constraints for legal admissibility of chain of custody and case records.
MongoDB: Manages unpredictable, heterogeneous surveillance telemetry (CCTV logs, phone extractions, and PDF forensic extracts) arriving without fixed schemas. Provides compound weighted full-text indexes and aggregation pipeline capabilities without altering relational tables.

2. MongoDB Schema Validation & Compound Text Indexes

```
// Collection Validator (database/mongodb/collection-schema.json)
{
  "$jsonSchema": {
    "bsonType": "object",
    "required": ["id", "title", "document_type", "source_name", "content_text", "ingestion_timestamp"],
    "properties": {
      "id": { "bsonType": "string" },
      "case_id": { "bsonType": ["string", "null"] },
      "document_type": { "enum": ["Witness Statement", "CCTV Log", "Phone Extraction", "Email Record", "Forensic Report"] },
      "source_name": { "bsonType": "string" },
      "content_text": { "bsonType": "string" },
      "tags": { "bsonType": "array", "items": { "bsonType": "string" } },
      "metadata": { "bsonType": "object" }
    }
  }
}

// Weighted Text Index Definition (database/mongodb/indexes.js)
db.investigation_documents.createIndex(
  { title: "text", tags: "text", content_text: "text" },
  { weights: { title: 10, tags: 5, content_text: 1 }, name: "idx_text_search" }
);
```

3. MongoDB Aggregation Pipeline Example

```
// Aggregation: Group intelligence artifacts by source type and compute average word count
db.investigation_documents.aggregate([
  { $match: { case_id: { $ne: null } } },
  { $group: {
      _id: "$document_type",
      total_docs: { $sum: 1 },
      sources: { $addToSet: "$source_name" },
      avg_content_len: { $avg: { $strLenCP: "$content_text" } }
  }},
  { $sort: { total_docs: -1 } }
]);
```